

MAT242 - Topology

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Intro

The following are my notes for Topology which I will (hopefully) be taking in the fall of 2026. As these are my own personal notes I may skip certain topics I already understand, be sloppy at times and this may cause trouble for anyone who wishes to use these notes for themselves so keep that in mind if you want to use these.

1 Topological spaces and continuity

We recall from Real Analysis that for a metric space (X, d) we have a notion of distance and can start describing open sets $U \subseteq X$ as sets where for every $x \in U$ there exists $\varepsilon > 0$ such that

$$B(x; \varepsilon) \subseteq U,$$

meaning that for any x in U there is some open ball around it contained entirely in U .

In Hilbert spaces, normed spaces, $\mathbb{R}^n$ (different types and instances of metric spaces), etc., this is the usual idea: an open set is one where every point has some wiggle room around it.

Topology starts from the observation that many arguments in Analysis don't really use the distance $d(x, y)$. They only use collections of open sets.

So instead of letting open sets be generated by some metric, why don't just specify which sets are open?

Intuitively, a topology on a set X is a choice about which sets are "open", whatever that may mean. Informally a topology tells us which points are "near" each other without really specifying quantitatively what that may mean.

So we want the least structure possible to be able to talk about things like continuity, connectedness, compactness, convergence, etc.

Definition 1.1 (Topological space). Let X be a set. A **topology** on X is a collection $\tau \subseteq \mathcal{P}(X)$ satisfying:

1. $\emptyset, X \in \tau$.
2. If $\{U_i\}_{i \in I}$ is any collection of elements in τ , then

$$\bigcup_{i \in I} U_i \in \tau.$$

3. If $U_1, \dots, U_n \in \tau$, then

$$U_1 \cap \dots \cap U_n \in \tau.$$

The pair (X, τ) is called a **topological space**. The elements of τ are called **open sets**.

We will often refer to a topological space by just the set if it is clear what the topology on X is.

The axioms are not arbitrary, but abstracted from metric spaces. In metric spaces we already have these properties even if they are not explicitly required. For instance in $\mathbb{R}$,

$$\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}.$$

Each interval is open, but $\{0\}$ is not.

Intuitively, finite intersections preserve some positive amount of wiggle room while infinite intersections can squeeze it all away.

Examples of topological spaces

Example 1.1 ($\mathbb{R}$). Let $X = \mathbb{R}$. The usual topology is the collection of subsets $U \subseteq \mathbb{R}$ such that for every $x \in U$, there exists $\varepsilon > 0$ with

$$(x - \varepsilon, x + \varepsilon) \subseteq U.$$

This is the topology we know from Real analysis. Examples of open sets include:

$$(0, 1), (-3, 7), \mathbb{R}, \emptyset,$$

and also things like

$$(0, 1) \cup (2, 5) \cup (10, \infty).$$

Non-examples:

$$[0, 1], \mathbb{Q}, \{0\}.$$

Note that $\mathbb{Q}$ can be made into a topological space by giving it the subspace topology from $\mathbb{R}$. More on that later.

Example 1.2 (Metric spaces). Every metric space (X, d) gives a topology. Define τ_d by saying that $U \subseteq X$ is open if for every $x \in U$ there exists $\varepsilon > 0$ such that

$$B(x; \varepsilon) \subseteq U.$$

Then τ_d is a topology on X . So every metric space is a topological space, but note that the opposite implication does not hold which is one of the reasons topology is more general.

Example 1.3 (The discrete topology). Let X be any set. Let $\tau := \mathcal{P}(X)$. Then τ is a topology on X which we call the **discrete topology**.

Example 1.4 (The indiscrete topology). Let X be any set. Define $\tau = \{\emptyset, X\}$. This is called the **indiscrete topology** or **trivial topology**.

Definition 1.2 (Closed set). A subset $C \subseteq X$ is called **closed** if its complement is open:

$$C \text{ is closed} \Leftrightarrow X \setminus C \text{ is open.}$$

In a topological space, $\emptyset$ and X are closed. Arbitrary intersections of closed sets are closed and finite unions of closed sets are closed.

An important nuance is that closed does not mean not open. $\emptyset, X$ are both open and closed. In $\mathbb{R}$ (with the usual topology) we have that $\emptyset, \mathbb{R}$ are both open and closed while e.g. $[0, 1)$ is neither open nor closed.

Continuity

In Analysis, a function on metric spaces $f : X \rightarrow Y$ is continuous at $x \in X$ if small changes in x in $f(x) \in Y$. Recall the ε - δ definition:

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t. } d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon.$$

Since we don't have access to metrics we must rewrite this.

Definition 1.3 (Continuity). Let (X, τ_X) and (Y, τ_Y) be topological spaces. A function

$$f : X \rightarrow Y$$

is **continuous** if for every open set $V \in \tau_Y$, the preimage

$$f^{-1}(V)$$

is open in X .

Proposition 1.1. Let (X, d_X) and (Y, d_Y) be metric spaces. A function

$$f : X \rightarrow Y$$

is continuous in the ε - δ sense if and only if for every open set $V \subseteq Y$, the preimage

$$f^{-1}(V)$$

is open in X .

Proof. First suppose f is continuous in the ε - δ sense. Let $V \subseteq Y$ be open. We want to show that $f^{-1}(V)$ is open in X . Take $x \in f^{-1}(V)$.

Then $f(x) \in V$. Since V is open, there exists $\varepsilon > 0$ such that

$$B(f(x); \varepsilon) \subseteq V.$$

By continuity of f at x , there exists $\delta > 0$ such that

$$d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon.$$

Thus, if $y \in B(x; \delta)$, then

$$f(y) \in B(f(x); \varepsilon) \subseteq V.$$

Therefore

$$y \in f^{-1}(V).$$

So

$$B(x; \delta) \subseteq f^{-1}(V).$$

Since every point of $f^{-1}(V)$ has a ball around it contained in $f^{-1}(V)$, the set $f^{-1}(V)$ is open.

Now suppose that for every open set $V \subseteq Y$, the preimage $f^{-1}(V)$ is open in X . Let $x \in X$, and let $\varepsilon > 0$. The ball

$$B(f(x); \varepsilon)$$

is open in Y . Therefore

$$f^{-1}(B(f(x); \varepsilon))$$

is open in X . Since $x \in f^{-1}(B(f(x); \varepsilon))$, there exists $\delta > 0$ such that

$$B(x; \delta) \subseteq f^{-1}(B(f(x); \varepsilon)).$$

Thus, if

$$d_X(x, y) < \delta,$$

then

$$y \in f^{-1}(B(f(x); \varepsilon)),$$

so

$$f(y) \in B(f(x); \varepsilon).$$

Equivalently,

$$d_Y(f(x), f(y)) < \varepsilon.$$

Hence f is continuous at x . Since x was arbitrary, f is continuous. $\square$

Suppose τ_1 and τ_2 are two topologies on the same set X .

We say τ_1 is **finer** than τ_2 if

$$\tau_2 \subseteq \tau_1.$$

That means τ_1 has more open sets. Equivalently, τ_2 is **coarser** than τ_1 . For every set X :

$$\{\emptyset, X\} \subseteq \tau \subseteq \mathcal{P}(X).$$

So the indiscrete topology is the coarsest topology possible, and the discrete topology is the finest topology possible.

This affects continuity. If the domain topology is finer, continuity is easier. If the target topology is finer, continuity is harder.

Example 1.5. Let X have the discrete topology and let Y be any topological space. Then every $f: X \rightarrow Y$ is continuous. This is because $f^{-1}(V) \subseteq X$ and every subset of X is open.

Example 1.6. Let Y have the indiscrete topology. Then every function $f: X \rightarrow Y$ is continuous.

The preimages

$$f^{-1}(\emptyset) = \emptyset, \quad f^{-1}(Y) = X$$

are both open in X .

Example 1.7. Let X be a set with two topologies τ_1, τ_2 . Then $\text{id}: (X, \tau_1) \rightarrow (X, \tau_2)$ is continuous exactly when $\tau_2 \subseteq \tau_1$.

First glimpse of homeomorphisms

Topology is about spaces up to continuous deformation. The correct notion of "same space" is not equality, but **homeomorphism**.

Definition 1.4 (Homeomorphism). Let X, Y be topological spaces. A function $X \rightarrow Y$ is a **homeomorphism** if

1. f is bijective,
2. f is continuous,
3. f^{-1} is continuous.

If such an f exists, then X and Y are homeomorphic.

Intuitively, homeomorphic spaces have the same topological shape. Think about those videos of coffee cups and donuts. This is what they meant.

Example 1.8. $(0, 1)$ and $\mathbb{R}$ are homeomorphic. One possible homeomorphism is

$$f: (0, 1) \rightarrow \mathbb{R}, \quad x \mapsto \tan\left(\pi x - \frac{\pi}{2}\right).$$

So $\mathbb{R}$ is homeomorphic to $(0, 1)$ even though $(0, 1)$ is "small" (bounded) in comparison to $\mathbb{R}$.

Lastly we remark that openness doesn't need to have anything to do with distance. In algebraic geometry and commutative algebra we look at the prime spectrum of a ring,

$$\text{Spec}(A)$$

which we endow with a topology called the **Zariski topology**. Closed sets are defined by ideals:

$$V(I) = \{\mathfrak{p} \in \text{Spec}(A) : I \subseteq \mathfrak{p}\}.$$

So the closed subsets are controlled by equations and ideals.

Exercises

Problem 1.1. Let X be any set. Prove that the discrete topology $\tau = \mathcal{P}(X)$ is a topology on X .

Proof. $\tau \subseteq \mathcal{P}(X)$ is certainly satisfied. Furthermore $\emptyset, X \in \tau = \mathcal{P}(X)$ is also satisfied.

Now let I be some index set. Since $\mathcal{P}(X)$ contains all subsets of X and we necessarily have

$$\bigcup_{i \in I} U_i \subseteq X$$

for $U_i \in \mathcal{P}(X)$, and then we have

$$\bigcup_{i \in I} U_i \in \mathcal{P}(X) = \tau.$$

Thus τ is closed under arbitrary intersections.

Let $U_1, \dots, U_n \in \tau = \mathcal{P}(X)$. Once again since $U_i \subseteq X$ for all $1 \leq i \leq n$ we have

$$\bigcap_{i \in [n]} U_i \subseteq X$$

so

$$\bigcap_{i \in [n]} U_i \in \tau = \mathcal{P}(X).$$

□

Problem 1.2. Let X be any set. Prove that $\tau = \{\emptyset, X\}$ is a topology on X .

Proof. We have $\emptyset, X \in \tau$ by definition and also $\tau \subseteq \mathcal{P}(X)$.

Now notice that the only non-trivial collection of open sets in τ is the entirety of τ :

$$\emptyset, X,$$

and this collection is finite. Then $\emptyset \cup X = X \in \tau$ and $\emptyset \cap X = \emptyset \in \tau$.

Thus τ is closed under arbitrary unions and finite intersection and hence forms a topology on X . □

Problem 1.3. Let X be a topological space. Prove that arbitrary intersections of closed sets are closed.

Proof. Let I be some index set so that C_i is closed for every $i \in I$. Then $X \setminus C_i$ is open for every $i \in I$. Then

$$\bigcup_{i \in I} X \setminus C_i = \bigcup_{i \in I} X \cap C_i^c$$

is open. Taking the complement we get

$$\left(\bigcup_{i \in I} X \cap C_i^c \right)^c = \bigcap_{i \in I} \emptyset \cup C_i = \bigcap_{i \in I} C_i$$

so

$$\bigcap_{i \in I} C_i$$

is closed. □

Problem 1.4. Let τ_1, τ_2 be topologies on X . Prove that $\text{id} : (X, \tau_1) \rightarrow (X, \tau_2)$ is continuous if and only if $\tau_2 \subseteq \tau_1$.

Proof. Suppose $\tau_2 \subseteq \tau_1$. Let $U \in \tau_2$. Then $U \in \tau_1$ and

$$\text{id}^{-1}(U) = U \in \tau_1.$$

Thus the preimage of $U \in \tau_2$ is in τ_1 . As U was arbitrary we have that id is continuous.

Suppose id is continuous. Then for any open set $U \in \tau_2$ we have that

$$\text{id}^{-1}(U) = U \in \tau_1$$

hence

$$\tau_2 \subseteq \tau_1.$$

□

1.1 Some exercises from Section 13 of Munkres

Problem 1.5 (Problem 1). Let X be a topological space; let A be a subspace of X . Suppose that for each $x \in A$ there is an open set U containing x such that $U \subset A$. Show that A is open in X .

Proof. For each $x \in A$, let U_x denote the open set containing x such that $U_x \subset A$. We claim that

$$A = \bigcup_{x \in A} U_x.$$

For each $x \in A$, $U_x \subset A$ so

$$\bigcup_{x \in A} U_x \subseteq A.$$

Now, for some arbitrary $x \in A$, there exists an open set U_x such that $x \in U_x$. Thus

$$A \subseteq \bigcup_{x \in A} U_x.$$

The two inclusions give the result. $\square$

Problem 1.6 (Problem 3). Show that the collection $\mathcal{T}_c$ given in Example 4 of §12 (of Munkres) is a topology on the set X . Is the collection

$$\mathcal{T}_\infty = \{U \mid X \setminus U \text{ is infinite or empty or all of } X\}$$

a topology on X ?

Note: For a set X , define $\mathcal{T}_c$ to be the collection of all subsets U of X such that $X \setminus U$ is either countable or all of X .

Proof. Since $X \setminus \emptyset = X$ so $\emptyset \in \mathcal{T}_c$. Furthermore, since $X - X = \emptyset$ which is finite and thus countable, we have $X \in \mathcal{T}_c$.

Let $\{U_\alpha\}$ be a family of open sets (ignoring the trivial case where $U_\alpha = \emptyset$ for every such U_α) in $\mathcal{T}_c$. By definition, for each U , we have that

$$|X \setminus U_\alpha| \leq \aleph_0.$$

Since

$$\left| \bigcup_{\alpha} U_\alpha \right| \geq |U_\alpha|$$

for any α , we have

$$\left| X \setminus \bigcup_{\alpha} U_\alpha \right| \leq |X \setminus U_\alpha| \leq \aleph_0.$$

Hence $\bigcup_{\alpha} U_\alpha \in \mathcal{T}_c$ so $\mathcal{T}_c$ is closed under arbitrary unions.

Now let $\{U_i\}_{i \in [n]}$ be a finite family of open sets. Assume also that $U_i \neq \emptyset$ for every i as that case is trivial. Then

$$X \setminus \bigcap_{i=1}^n U_i = \bigcup_{i=1}^n (X \setminus U_i).$$

By assumption, $X \setminus U_i$ is countable and since the union of countable sets is countable we have that $\mathcal{T}_c$ is closed under finite intersections.

Now to answer the second question, $\mathcal{T}_\infty$ is absolutely not a topology on X . Take for instance $X = \mathbb{N}$. Then $\{n\} \in \mathcal{T}_\infty$, but

$$\mathbb{N} \setminus \bigcup_{n \in \mathbb{N} \setminus \{7\}} \{n\}$$

is $\{7\}$ which is neither infinite, empty, or all of X . $\square$

Problem 1.7 (Problem 4).

a) If $\{\mathcal{T}_\alpha\}$ is a family of topologies on X , show that $\bigcup \mathcal{T}_\alpha$ is a topology on X . Is $\bigcap \mathcal{T}_\alpha$ a topology on X ?

Proof of (a). Suppose $\{\mathcal{T}_\alpha\}$ is a family of topologies on X . Then $\emptyset, X \in \mathcal{T}_\alpha$ for all α so

$$\emptyset, X \in \bigcap \mathcal{T}_\alpha.$$

Let $\{U_\beta\}$ be a family of open sets in $\bigcap \mathcal{T}_\alpha$. Then every U_β lies in $\mathcal{T}_\alpha$ for every α . Thus

$$\bigcup U_\beta \in \mathcal{T}_\alpha, \quad \forall \alpha$$

and hence $\bigcup U_\beta$ lies in the intersection.

Now let $\{U_i\}_{i \in [n]}$ be a finite family of open sets in $\bigcap \mathcal{T}_\alpha$.

Again, $U_i \in \mathcal{T}_\alpha$ for every i and α and the argument is analogous to the one above.

The same does not hold for $\bigcup \mathcal{T}_\alpha$. Suppose $X = \{1, 2, 3\}$ with $\mathcal{T}_1 = \{\emptyset, X, \{1\}\}$ and $\mathcal{T}_2 = \{\emptyset, X, \{2\}\}$. Then $\{1\}$ and $\{2\}$ lie in their union, but $\{1, 2\}$ does not. $\square$

Problem 1.8 (Problem 6). Show that the topologies $\mathbb{R}_\ell$ and $\mathbb{R}_K$ are not comparable.

Proof. Recall that the topology of $\mathbb{R}_\ell$ is

$$\mathcal{B}_\ell = \{[a, b) \mid a < b\},$$

and that the topology of $\mathbb{R}_K$ is

$$\mathcal{B}_K = \{(a, b) : a < b\} \cup \{(a, b) \setminus K : a < b\}$$

where

$$K = \left\{ \frac{1}{n} \mid n \in \mathbb{Z}_+ \right\}.$$

These are not comparable. Take for instance $(-1,1) \setminus K \in \mathcal{B}_K$. If $\mathcal{B}_K \leq \mathcal{B}_\ell$ we should be able to find some $\varepsilon > 0$ so that $[0,\varepsilon) \subseteq (-1,1) \setminus K$, but for every $\varepsilon > 0$ we can choose n sufficiently large so that

$$\frac{1}{n} < \varepsilon.$$

Then $1/n \in [0,\varepsilon)$ implying $[0,\varepsilon) \not\subseteq (-1,1) \setminus K$. Thus $\mathcal{B}_K \not\subseteq \mathcal{B}_\ell$

For the other way consider $[0,1) \in \mathcal{B}_\ell$. Every K -neighbourhood at 0 is either (a,b) or $(a,b) \setminus K$ with $a < 0 < b$. Either way it contains some negative real numbers so no such basic neighborhood of 0 can fit in $[0,1)$. $\square$

Problem 1.9 (Problem 8).

a) Apply Lemma 13.2 to show that the countable collection

$$\mathcal{B} = \{(a,b) \mid a < b, a,b \in \mathbb{Q}\}$$

is a basis that generates the standard topology on $\mathbb{R}$.

Proof of (a). Recall that Lemma 13.2 says that if X is a topological space, and $\mathcal{C}$ is a collection of open sets such that for each open set U of X and each $x \in U$ there is some element $C \in \mathcal{C}$ such that $x \in C \subset U$, then $\mathcal{C}$ is a basis for the topology on X .

Let U be open in the canonical topology on $\mathbb{R}$. Then $U = (a_0, b_0)$ for some $a_0, b_0 \in \mathbb{R}$ such that $a_0 < b_0$. By virtue of $\mathbb{Q}$ being dense in $\mathbb{R}$ we can, for each $\varepsilon > 0$ find some $a, b \in \mathbb{Q}$ suitably close:

$$a - a_0 < \varepsilon, \quad b_0 - b < \varepsilon.$$

For any $x \in U = (a_0, b_0)$ let $\varepsilon = \min\{x - a_0, b_0 - x\} > 0$. Choose rational a and b (with $a < b$) so that $a - a_0, b_0 - b < \varepsilon$. Then

$$\begin{aligned} a - a_0 &< \varepsilon \\ &\leq x - a_0 \\ a &< x, \end{aligned}$$

and

$$\begin{aligned} b_0 - b &< \varepsilon \\ &\leq b_0 - x \\ -b &\leq -x \\ b &> x. \end{aligned}$$

Thus, by Lemma 13.2, $\mathcal{B}$ is a basis for the standard $\mathbb{R}$ topology.

This is a bit imprecise in places, but the idea is right. $\square$

2 Some problems to recap

Problem 2.1 (Munkres §16, Exercise 3). Let

$$[-1, 1] \subseteq \mathbb{R}$$

with the subspace topology inherited from the standard topology on the reals.

For each of the following sets, determine:

1. Is it open in Y ?

2. Is it open in $\mathbb{R}$?

$$\bullet A = \left\{ x \mid \frac{1}{2} < |x| < 1 \right\},$$

$$\bullet B = \left\{ x \mid \frac{1}{2} < |x| \leq 1 \right\},$$

$$\bullet C = \left\{ x \mid \frac{1}{2} \leq |x| < 1 \right\},$$

$$\bullet D = \left\{ x \mid \frac{1}{2} \leq |x| \leq 1 \right\},$$

and

$$\bullet E = \{ x \mid 0 < |x| < 1 \text{ and } 1/x \notin \mathbb{Z}_+ \}.$$

Sol. A set X is open in $\mathbb{R}$ if for every $x \in X$, there is some $\varepsilon > 0$ such that $(x - \varepsilon, x + \varepsilon) \subseteq X$. This is the same behavior inherited by Y .

We begin with A . We claim that this set is indeed open.

We have that

$$\frac{1}{2} < x < 1 \text{ or } -1 < x < -\frac{1}{2}.$$

Hence $A = (-1, -1/2) \cup (1/2, 1)$. The union of open sets is open hence A is open.

Now we do B . We claim that this set is not open. Let $x = 1$. There would have to exist $\varepsilon > 0$ so that $(x - \varepsilon, x + \varepsilon) \subseteq X$, but $x + \varepsilon > 1$ and hence not in B .

Now we do C . We also claim this set is not open using the same argument. An open ε -ball of $1/2$ would include $1 - \varepsilon/2 < 1/2$.

D fails for the same reason.

$$E = (-1, 0) \cup \bigcup_{n=1}^{\infty} \left(\frac{1}{n+1}, \frac{1}{n} \right),$$

a union of open sets, hence E is open. □

Problem 2.2. Let

$$f : X \rightarrow Y$$

and suppose $\mathcal{B}$ is the basis for the topology on Y .
Prove that f is continuous if and only if

$$f^{-1}(B)$$

is open for every $B \in \mathcal{B}$.

Proof. Suppose f is continuous. Since $B \in \mathcal{B}$ are necessarily open in the topology they induce it is trivial that

$$f^{-1}(B)$$

is open, by definition.

Now suppose

$$f^{-1}(B) = \{x \in X \mid f(x) \in B\}$$

is open for every $B \in \mathcal{B}$. Recall that every open set $U \subseteq Y$ can be written as

$$U = \bigcup_{\alpha} B_{\alpha}, \quad B_{\alpha} \in \mathcal{B}.$$

Then

$$f^{-1}\left(\bigcup_{\alpha} B_{\alpha}\right) = \{x \in X \mid x \in f^{-1}(B_{\alpha}) \text{ for at least one } \alpha\}.$$

In particular

$$f^{-1}\left(\bigcup_{\alpha} B_{\alpha}\right) = \bigcup_{\alpha} f^{-1}(B_{\alpha})$$

and the preimage of B_{α} is, by assumption, open in X . As topologies are closed under arbitrary unions we get that $f^{-1}(U)$ is open in X . As U was arbitrary, this holds for any open $U \subseteq Y$. □

Problem 2.3. Let X, Y, Z be topological spaces and let

$$f : Z \rightarrow X \times Y.$$

Write

$$f(z) = (f_1(z), f_2(z))$$

where f_1, f_2 are the projections to coordinates 1 and 2 respectively.
 Prove that f is continuous if and only if f_1, f_2 are continuous.

Proof. The basis of the product topology is

$$\{U \times V \mid U \subseteq X \text{ is open, and } V \subseteq Y \text{ is open}\}.$$

Suppose f is continuous. Then

$$f_1 = \pi_1 \circ f, \text{ and } f_2 = \pi_2 \circ f$$

where π_1, π_2 are projections. Thus f_1, f_2 are compositions of continuous functions which is continuous.

Now suppose f_1, f_2 are continuous.

An open set $W \subseteq Y$ can be written as

$$W = \bigcup_{\alpha} U_{\alpha} \times V_{\alpha}$$

Then $f^{-1}(W)$ is the set of all $z \in Z$ such that $f_1(z) \in U_{\alpha}$ for at least one α and $f_2(z) \in V_{\alpha}$ for at least one alpha so

$$f^{-1}(W) = \bigcup_{\alpha} f_1^{-1}(U_{\alpha}) \cap f_2^{-1}(V_{\alpha}).$$

As the $f_1^{-1}(U_{\alpha}), f_2^{-1}(V_{\alpha})$ are open in Z , their finite union is open in Z and hence the union of $f_1^{-1}(U_{\alpha}), f_2^{-1}(V_{\alpha})$ over α is open in Z . $\square$

Problem 2.4. Let $\mathcal{T}, \mathcal{T}'$ be topologies on a set X with $\mathcal{T} \subseteq \mathcal{T}'$, meaning $\mathcal{T}$ is finer than $\mathcal{T}'$. Considering the identity map

$$\text{id} : X \rightarrow X$$

we can regard it in two ways, the one where $\mathcal{T}'$ is carried to $\mathcal{T}$ or the other way. For which of these is the identity map always continuous?

Proof. The map

$$\text{id} : (X, \mathcal{T}') \rightarrow (X, \mathcal{T})$$

is always continuous, the other doesn't need to be.

Let $U \in \mathcal{T}$ be open. Then

$$\text{id}^{-1}(U) = U \in \mathcal{T}'.$$

Consider $\mathcal{T} = \{\emptyset, X\} \subseteq \mathcal{T}' = \{\emptyset, X, A\}$. Then the other identity map has

$$\text{id}(A)^{-1} = A \notin \mathcal{T}.$$

$\square$

3 Quotient spaces and quotient topology

Quotient topology will be in spirit antithetical to the product topology. The fundamental question is: if we glue points of some space X together, what topology should the space have?

We begin by recapping what equivalence relations are.

Definition 3.1. Let X be a set.

An equivalence relation, on X is relation $\sim$ satisfying reflexivity:

$$x \sim x$$

for every $x \in X$,
symmetry:

$$x \sim y \Leftrightarrow y \sim x,$$

and
transitivity:

$$x \sim y, y \sim z \Rightarrow x \sim z.$$

For $x \in X$ we denote the equivalence class

$$[x] = \{y \in X \mid y \sim x\}.$$

Equivalence classes partition X .

The quotient set is

$$X / \sim = \{[x] \mid x \in X\}.$$

There is a canonical map

$$q : X \rightarrow X / \sim, \quad q(x) = [x].$$

This map is automatically surjective.

This is a nice construction in topology because these relations give us a sort of language for gluing.

Take $X = [0, 1]$. Suppose we identify two endpoints $0 \sim 1$, while every other endpoint is equivalent to only itself. Then the quotient

$$[0, 1] / \sim$$

is like a circle.

Likewise starting from a square

$$[0, 1] \times [0, 1]$$

and identifying the left and right edges appropriately gives a cylinder.

Now the natural question is what topology should be endowed on this space?

Suppose X is a topological space and let

$$q : X \rightarrow Y$$

be a surjection.

We want to make Y a topological space.

Since q tells us which points of X collapse onto the same point in Y , the natural idea is a set downstairs should be open exactly when its full preimage upstairs is open.

Definition 3.2 (Quotient topology). *The quotient topology on Y is*

$$\tau_Y = \{U \subseteq Y \mid q^{-1}(U) \text{ open in } X\}.$$

Proposition 3.1. *The quotient topology is indeed a topology.*

Proof. First,

$$q^{-1}(\emptyset), q^{-1}(Y) = X.$$

Both are open in X .

Let $\{U_\alpha\}_\alpha \subseteq \tau_Y$. Then

$$q^{-1}\left(\bigcup_\alpha U_\alpha\right) = \bigcup_\alpha q^{-1}(U_\alpha).$$

Each $q^{-1}(U_\alpha)$ is open in X so the union of these is open in X .

If $U_1, \dots, U_n \in \tau_Y$, then

$$q^{-1}\left(\bigcup_{i=1}^n U_i\right) = \bigcap_{i=1}^n q^{-1}(U_i)$$

which is the finite intersection of open sets in the domain and thus open. $\square$

We need the map to be surjective otherwise there could be points in the codomain which are never seen.

A surjective continuous map $q : X \rightarrow Y$ is called a quotient map if $U \subseteq Y$ if and only if $q^{-1}(U)$ is open in X .

A quotient space, on the other hand, is typically a space

$$X/\sim$$

equipped with the quotient topology induced by the canonical projection $q : X \rightarrow X/\sim$.

It is tempting to think that

$$U \subseteq X \text{ open} \Rightarrow q(U) \subseteq Y \text{ open}.$$

This is false in general.

The definition works with the preimage not with the image. Just like how topology in general behaves quite nicely with preimages.

Definition 3.3. A subset $A \subseteq X$ is called saturated with respect to q if it contains every fibre that it touches. Formally,

$$A = q^{-1}(q(A)).$$

In the case of an equivalence relation this means

If $x \in A$, then every point is equivalent to x is also in A .

Every set of the form $q^{-1}(A)$ is saturated.

Thus the quotient topology is really controlled by open saturated subsets of X .

Example 3.1. Let X be the closed unit interval. Define $0 \sim 1$ with no other nontrivial identifications. Then $Y = X / \sim$ has one special point

$$[0] = [1].$$

Suppose $U \subseteq Y$ contains $[0]$. Its preimage must contain both 0 and 1. For U to be open in Y it must be open in the subspace $[0, 1]$. So it must contain something like

$$[0, \varepsilon) \cup (1 - \varepsilon, 1].$$

After quotienting, these two pieces become one neighborhood around the glued point. That is exactly what neighborhoods around a point on the circle look like.

Now we arrive at the universal property.

Suppose

$$q : X \rightarrow Y$$

is a quotient map.

Let Z be another topological space and

$$f : X \rightarrow Z.$$

Then

$$f \text{ is continuous} \Leftrightarrow f \circ q \text{ is continuous.}$$

You can illustrate the difference between this- and the product topology's universal property particularly nice with diagrams in a category theory-esque way, but I'm lazy.

Now we pose a natural question.

Suppose

$$q : X \rightarrow X / \sim$$

is the quotient map. Imagine we have a continuous function

$$g: X \rightarrow Z.$$

When can it descend to a function $\bar{g}: X/\sim \rightarrow Z$ such that $g = \bar{g} \circ q$?

Set-theoretically, this is possible exactly when g is constant on equivalence classes:

$$x \sim y \Rightarrow g(x) = g(y).$$

If that happens, define

$$\bar{g}([x]) = g(x).$$

Because g is constant on equivalence classes, this is well-defined.

Then the quotient universal property gives

$$g = \bar{g} \circ q$$

continuous

$$\Rightarrow \bar{g} \text{ continuous.}$$

So quotient spaces let you convert functions on X that respect the identification into functions on the glued space.

This is perhaps the most useful operational idea:

If a continuous map cannot distinguish points you glue together,
it descends continuously to the quotient.

Problem 3.1. Let $X = [0, 1]$ and define $0 \sim 1$ with all other equivalence classes singletons. Let

$$q: X \rightarrow X/\sim$$

be the quotient map. Consider

$$A = q([0, 1/4) \cup (3/4, 1]).$$

Try to determine whether A is open in the quotient space.

Sol. Let $B = [0, 1/4) \cup (3/4, 1]$. The forward inclusion

$$B \subseteq q^{-1}(q(B))$$

is always true. Indeed, if $x \in B$ then $q(x) \in q(B)$ so

$$x \in q^{-1}(q(B)).$$

For the reverse, suppose $x \in q^{-1}(q(B))$. Then

$$q(x) \in q(B),$$

so there exists $b \in B$ such that

$$q(x) = q(b).$$

But $q(x) = q(b)$ means that

$$x \sim b.$$

The only nontrivial class is $\{0,1\}$. Since B contains both 0,1 whenever $b \in B$ every point is equivalent to b in B so $x \in B$.

The two inclusions mean that B is saturated under the equivalence relation. Hence A is open in the quotient space. $\square$

Problem 3.2.

- a) Let $p : X \rightarrow Y$ be a continuous map. Show that if there is a continuous map $f : Y \rightarrow X$ such that $p \circ f$ equals the identity map of Y , then p is a quotient map.
- b) If $A \subset X$, a retraction of X onto A is a continuous map $r : X \rightarrow A$ such that $r(a) = a$ for each $a \in A$. Show that a retraction is a quotient map.

Proof of (a). Recall that a quotient map $g : X \rightarrow Y$ is a surjection such that $U \subseteq Y$ is open if and only if $g^{-1}(U)$ is open.

Condition two is partially satisfied by p since, by assumption, $p^{-1}(U)$ is open when $U \subseteq Y$ is open.

Suppose p is not a surjection. Then there are $y \in Y$ such that, for every $x \in X$,

$$p(x) \neq y.$$

Then, $p \circ f \neq \text{id}_Y$. $\text{im } f \subseteq X$ so there is no $x \in \text{im } f$ so that

$$p(x) = y.$$

In particular, $f(y) \in \text{im } f \subseteq X$ so $f(y) \neq y$.

Thus we require that p is a surjection.

Now it remains to show that p carries open sets to open sets. Suppose $V \subseteq X$ is open. Then since f is continuous, $f^{-1}(V) \subseteq Y$ is open.

We have that

$$p \circ f(f^{-1}(V)) = f^{-1}(V)$$

so

$$p(V) = f^{-1}(V).$$

Thus p carries open sets to open sets.

Summing up, we have that p is a quotient map. $\square$

Proof of (b). Let $r : X \rightarrow A$ be a retraction, and let

$$i \hookrightarrow X$$

denote the inclusion map, defined by $i(a) = a$.

Since A has the subspace topology, i is continuous. Moreover, since r is a retraction,

$$(r \circ i)(a) = r(a) = a$$

for every $a \in A$. Hence

$$r \circ i = \text{id}_A.$$

Thus r has a continuous right inverse. By part *a*, r is a quotient map. $\square$

Problem 3.3.

a) Define an equivalence relation on the plane $X = \mathbb{R}^2$ as follows:

$$x_0 \times y_0 \sim x_1 \times y_1 \text{ if } x_0 + y_0^2 = x_1 + y_1^2.$$

Let X^ be the corresponding quotient space. It is homeomorphic to a familiar space; what is it?*

b) Repeat for the equivalence relation

$$(x_0, y_0) \sim (x_1, y_1) \text{ if } x_0^2 + y_0^2 = x_1^2 + y_1^2.$$

Solution to (a). Define $g([x, y]) : X^* \rightarrow \mathbb{R}$, $g([x, y]) = x + y^2$.

Then,

$$(x_0, y_0) \sim (x_1, y_1) \Leftrightarrow g([x_0, y_0]) = g([x_1, y_1]).$$

Thus the equivalence classes are the fibres of g .

We claim this map is the continuous inverse of the quotient map $q : \mathbb{R}^2 \rightarrow \mathbb{R}$.

Well-definedness of the map is immediate.

It is also injective immediately from the definition.

For surjectivity, given $t \in \mathbb{R}$, simply take $(t, 0)$. Then $g([t, 0]) = t$.

Notice $x + y^2 = g \circ q$ is continuous. By the defining property of the quotient topology, g is continuous exactly when $g \circ q$ is. Thus g is continuous.

Define $s : \mathbb{R} \rightarrow \mathbb{R}^2$ by $s(t) = (t, 0)$. Then s is continuous.

Now $(q \circ s)(t) = [(t, 0)]$, but this is exactly the inverse of g . So g^{-1} is continuous.

Thus $X^* \cong \mathbb{R}$. $\square$

Solution sketch of (b). Proceed as above and define $g : \mathbb{R}^2 \rightarrow [0, \infty)$ by $g(x, y) = x^2 + y^2$ to get $X^* \cong [0, \infty)$. $\square$

3.1 Some more problems

Problem 3.4. Let $q : X \rightarrow Y$ be a surjection. Prove that for every $A \subseteq X$,

$$A \subseteq q^{-1}(q(A)).$$

Show that equality holds if and only if A is a union of fibers of q . In the equivalence-relation setting, formulate this as " A is a union of equivalence classes".

Proof. Let $A \subseteq X$. Then $q(A) \subseteq Y$. We look at the preimage

$$q^{-1}(q(A)) = \{x \in X \mid q(x) \in q(A)\}.$$

Clearly, this must include A . Otherwise there would be at least one $a \in A$ such that $q(a) \notin q(A)$, but $q(A) = \{q(a) \mid a \in A\}$ (Arguing about sets is ridiculous).

Suppose the preimage is exactly A , meaning A is saturated w.r.t q . Take any $a \in A$. Consider the fiber of q through a :

$$q^{-1}(\{q(a)\}).$$

We claim this entire fiber lies in A .

Indeed, if

$$x \in q^{-1}(\{q(a)\}),$$

then

$$q(x) = q(a).$$

Since $a \in A$,

$$q(a) \in q(A),$$

so $q(x) \in q(A)$.

Thus

$$x \in q^{-1}(q(A)).$$

By assumption, $q^{-1}(q(A)) = A$, so $x \in A$.

Conversely, suppose A is a union of fibers. Then it remains to prove that

$$q^{-1}(q(A)) \subseteq A.$$

Take $x \in q^{-1}(q(A))$. Then $q(x) \in q(A)$, so by definition of the image there is some $a \in A$ such that $q(x) = q(a)$. Thus x and a lie in the same fiber. Hence

$$q^{-1}(q(A)) \subseteq A.$$

$\square$

Problem 3.5. Let $X \subseteq \mathbb{R}$ be the closed unit interval with $0 \sim 1$ and define $g: [0, 1] \rightarrow \mathbb{R}^2$ as

$$g(t) = (\cos 2\pi t, \sin 2\pi t).$$

Show that g is constant on equivalence classes, so it induces a well-defined map

$$\bar{g}: X/\sim \rightarrow S^1.$$

Then use the quotient universal property to prove $\bar{g}$ is continuous.

Proof sketch. $\cos$ and $\sin$ are 2π -periodic so that

$$\cos(2\pi \cdot 0) = \cos(2\pi \cdot 1) \text{ and } \sin(2\pi \cdot 0) = \sin(2\pi \cdot 1).$$

This, together with the rest of the equivalence classes being singletons, makes g constant on all the equivalence classes. Thus $\bar{g}: X/\sim \rightarrow S^1$ is well-defined.

As this map is the composition of g (continuous) with the canonical projection $x \mapsto [x]$ then, by the universal property of quotient topology, $\bar{g}$ is continuous.

Furthermore $\bar{g}$ is invertible (I will not rigorously prove this) so

$$X/\sim \cong S^1$$

the 1-dimensional loop. □

Problem 3.6. Let $X = \mathbb{R}$ and identify all integers to a single point, leaving all nonintegers equivalent only to themselves. Give $X/\sim$ the quotient topology. Decide whether the quotient is Hausdorff.

Proof. For the quotient map $q: X \rightarrow X/\sim$ we have $A \subseteq X/\sim$ is open if and only if its preimage is open in X .

The Hausdorff property is that for any two distinct points x, y we can find open neighborhoods for each which are disjoint.

We split it into two cases.

Case 1: $[\mathbb{Z}]$ and a noninteger point

Let $x \notin \mathbb{Z}$. Since $\mathbb{Z}$ is closed and discrete in $\mathbb{R}$, we can choose $\varepsilon > 0$ such that

$$[x - \varepsilon, x + \varepsilon] \cap \mathbb{Z} = \emptyset.$$

Put

$$V = (x - \varepsilon/2, x + \varepsilon/2)$$

and

$$U = R/[x - \varepsilon, x + \varepsilon].$$

Then U and V are disjoint w.r.t q . Therefore

$$q^{-1}(q(U)) = U, \quad q^{-1}(q(V)) = V.$$

Since U, V are open, the quotient topology implies that $Q(U), Q(V)$ are open in $\mathbb{R}/\sim$. They are also disjoint, because U, V are disjoint saturated sets. Finally

$$[\mathbb{Z}] \in q(U)$$

and

$$[x] \in q(V).$$

Thus $[\mathbb{Z}], [x]$ have disjoint open neighborhoods.

The case for two nonintegers is easy from the typical topology on $\mathbb{R}$. $\square$

4 Connectedness, components, compactness

From §23 of Munkres recall that a space X is connected if we cannot write $X = U \sqcup V$ where U, V are disjoint, nonempty, open subsets of X .

Equivalently, X is connected if and only if the only clopen subsets of X are X and $\emptyset$.

Also, X connected and $f: X \rightarrow Y$ continuous implies $f(X)$ connected.

4.1 §24, Connected subsets of the real line

The main theorem we will be keeping in mind is $A \subseteq \mathbb{R}$ is connected if and only if A is an interval, which is perhaps unsurprising.

Interval should be understood in the broad order-theoretic sense, meaning

$$x, z \in A, x < y < z \Rightarrow y \in A.$$

Such sets are also called convex subsets of the ordered set $\mathbb{R}$ in some contexts. Munkres develops this more generally for linear continua later, I believe.

The claim now is that connected subsets must be intervals.

Suppose $A \subseteq \mathbb{R}$ is connected. Take $x, z \in A$, $x < z$. Suppose some y satisfies $x < y < z$, but does not belong to A .

Then

$$A \cap (-\infty, y)$$

and

$$A \cap (y, \infty)$$

are disjoint nonempty open subsets of A , and together they cover A .

They are nonempty because x is in the first and z is in the second.

Hence they give a separation of A , contradicting connectedness. Therefore every point between x and z lies in A . So A is an interval.

4.2 The intermediate Value Theorem in topology

Now suppose X is connected and $f: X \rightarrow \mathbb{R}$ is continuous. Because continuous images of connected spaces are connected we have that $f(X)$ is connected, but connected subsets of $\mathbb{R}$ are intervals.

Therefore if

$$f(a) < r < f(b),$$

then

$$r \in f(X).$$

So there exists $c \in X$ with $f(c) = r$.

4.3 Path connectedness

Path connectedness is a strictly stronger notion than connectedness.

A path from x to y in X is a continuous map

$$\gamma : [0, 1] \rightarrow X$$

such that

$$\gamma(0) = x, \quad \gamma(1) = y.$$

A space X is path connected if every pair $x, y \in X$ can be joined by a path. Intuitively, connected says we cannot split the space, while path connected means we can travel continuously between all distinct points.

Proposition 4.1. *Path connectedness implies connectedness.*

Proof. Suppose X is path connected, but disconnected. Then $X = U \cup V$ is a separation. Choose $x \in U$ and $y \in V$. There is a path $\gamma : [0, 1] \rightarrow X$ from x to y , but $[0, 1]$ is connected, so its continuous image $\gamma([0, 1])$ is connected. Yet this image intersects both U and V , which would separate it, a contradiction. $\square$

4.4 §25, Components: how a disconnected space splits

Connectedness is a yes/no question asking "Is the whole space connected?". Components are slightly more refined and instead ask "What are the largest connected chunks of the space?".

For $x \in X$ define the components of x to be the union of all connected subsets of X containing x .

Equivalently, define $x \sim y$ if there exists a connected subset of X containing both x and y . The equivalence classes are the components of X . So a component is a maximal connected subspace.

Suppose

$$C_x = \bigcup_{\alpha} A_{\alpha},$$

where each A_{α} is connected and contains x . Every A_{α} has the common point x .

The union of connected sets having a common point is connected.

Therefore C_x is connected. And since it contains every connected set containing x , it is maximal.

Furthermore, components partition the space.

Every point lies in its own component. Two components are either disjoint or equal. Suppose C_1, C_2 intersect. Then $C_1 \cup C_2$ is connected

because the two connected sets have a common point. However, both C_1, C_2 are maximal connected sets so $C_1 = C_2$.

Thus X is partitioned into connected pieces.

4.5 Components are closed

This is an important theorem: Every component of a space is closed.

The main ingredient in this is an important one, namely that A connected $\Rightarrow \overline{A}$ connected. If C is a component, then its closure $\overline{C}$ is connected, but $C \subseteq \overline{C}$ and C was maximal so $\overline{C} = C$. Hence it is closed.

4.6 Path components

Similarly, define $x \sim_p y$ if there exists a path joining x and y . The equivalence classes are called path components. Every path component lies inside a component because path connected $\Rightarrow$ connected.

So conceptually, path components $\subseteq$ components and they can be strictly smaller.

4.7 Local connectedness

A space can fail to be connected globally, but look connected locally.

A space X is locally connected at x if every neighborhood U of x contains a connected neighborhood V of x .

That is,

$$x \in V \subseteq U$$

with V connected and a neighborhood of x .

It is locally connected if this holds at every $x \in X$. Munkres gives an analogous definition for local path connectedness. For example, $\mathbb{R} \setminus \{0\}$ is not connected, but it is locally connected, meaning around every point you can choose a sufficiently small interval not crossing 0.

4.8 Local theorem

Munkres gives the useful characterization,

X is locally connected



every component of every open subset of X is open in X .

There is, of course, an analogous statement with path components and locally path connected spaces.

Now, why should this make sense?

Suppose X is locally connected, $U \subseteq X$ is open, and C is a component of U . Take $x \in C$.

Because U is a neighborhood of x , local connectedness gives a connected neighborhood V_x with $x \in V_x \subseteq U$.

Since V_x is connected and intersects C , maximality of C gives $V_x \subseteq C$. Do this for every $x \in C$. Then

$$C = \bigcup_{x \in C} V_x.$$

Hence C is open.

Also, X locally path connected $\Rightarrow$ components = path components.

4.9 §26, Compactness

Connectedness asks whether the space can be pulled apart, while compactness asks whether infinitely much local information can be reduced to finitely much information. Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be a collection of open sets in X .

It is an open cover of X if

$$X \subseteq \bigcup_{\alpha \in A} U_\alpha.$$

The space X is compact if every open cover has a finite subcover, meaning

$$X \subseteq \bigcup_{\alpha \in A} U_\alpha \Rightarrow X \subseteq U_{\alpha_1} \cup \dots \cup U_{\alpha_n}$$

for some finite choice of indices.

As a first example, recall from analysis that $[0,1]$ is compact, but $(0,1)$ is not. For example, $U_n = (1/n, 1)$ cannot be reduced to a finite cover.

4.10 Closed subsets of compact spaces

Theorem 4.1. X is compact and $A \subseteq X$ closed implies that A is compact.

Proof. Let $\{U_\alpha\}$ be an open cover of A . Because A is closed, $X \setminus A$ is open. Then $\{U_\alpha\} \cup \{X \setminus A\}$ covers X and can thus be reduced to a

finite cover. Discard $X \setminus A$ if it was chosen. The remaining finitely many U_α 's cover A . $\square$

4.11 Compact subsets of Hausdorff spaces are closed

The converse is generally not true.

In a Hausdorff space, $A \subseteq X$, A compact, X Hausdorff $\Rightarrow A$ closed.

Take $x \in X \setminus A$. For every $a \in A$, Hausdorffness gives disjoint open $U_a \ni a$ and $V_a \ni x$. The sets U_a cover A . Compactness gives finitely many $U_{a_1}, \dots, U_{a_n}$.

Let

$$V = V_{a_1} \cap \dots \cap V_{a_n}.$$

This is open, contains x , and is disjoint from $U_{a_1} \cup \dots \cup U_{a_n}$, which contains A . Thus $V \cap A = \emptyset$. So every point outside A has an open neighborhood not intersecting with A . Therefore $X \setminus A$ is open, hence A is closed.

4.12 Two nice theorems

Another nice theorem I'll state informally:

X compact, $f : X \rightarrow Y$ continuous $\Rightarrow f(X)$ compact.

Let $\{U_\alpha\}$ be an open cover of $f(X)$. Then $\{f^{-1}(U_\alpha)\}$ covers X . Compactness lets us reduce to finitely many and therefore we can find a finite cover of the image of f .

Theorem 4.2. Suppose $f : X \rightarrow Y$ is continuous and bijective with X compact and Y Hausdorff. Then f is a homeomorphism.

Proof sketch. Take a closed set $C \subseteq X$. Since X is compact, C is compact. Continuous images preserve that property so $f(C)$ is compact in Y . Since Y is Hausdorff, $f(C)$ is closed. Thus f maps closed sets to closed sets. Since f is a bijection, this implies f^{-1} is continuous. $\square$

4.13 Some problems

Problem 4.1 (Munkres 24.1).

a) Show that no two of the spaces $(0,1)$, $(0,1]$, $[0,1]$ are homeomorphic.

- b) Suppose there exist imbeddings $f : X \rightarrow Y$ and $g : Y \rightarrow X$. Show by means of an example that X, Y need not be homeomorphic.
- c) Show $\mathbb{R}^n$ and $\mathbb{R}$ are not homeomorphic if $n > 1$.

Proof of (a). Notice that $[0, 1]$ is compact while $(0, 1)$ is not. The latter is not since, if we take for instance the cover

$$\{(1/n, 1)\}_{n \in \mathbb{N}}$$

of $(0, 1)$ this cannot be reduced to a finite cover since for every n we can find some $m > n$ such that

$$0 < \underbrace{1/m}_{\in (0, 1)} < 1/n.$$

So $[0, 1]$ and $(0, 1)$ cannot be homeomorphic since continuous images would preserve compactness.

$[0, 1]$ and $(0, 1]$ cannot be homeomorphic for the same reason.

Lastly $(0, 1)$ is not homeomorphic to $(0, 1]$. Suppose they were. Let $f : (0, 1] \rightarrow (0, 1)$ be the corresponding homeomorphism. Then

$$f|_{(0, 1] \setminus \{1\}} : (0, 1) \rightarrow (0, 1) \setminus f(\{1\})$$

would be a homeomorphism. Let $x = f(1)$. For any point in $y \in (0, 1)$, its removal leaves

$$(0, 1) \setminus \{y\} = (0, y) \cup (y, 1)$$

and in particular

$$(0, 1) \setminus f(\{1\}) = (0, f(1)) \cup (f(1), 1)$$

so the space is disconnected, while $(0, 1] \setminus \{1\} = (0, 1)$ is connected, a contradiction. $\square$

Problem 4.2 (Munkres 25.4). Let X be locally path connected. Show that every connected open set in X is path connected.

Proof. Let $U \subseteq X$ be open and connected. We show that U is path connected.

Fix $x \in U$ and let

$$P_x = \{z \in U : \text{there exists a path in } U \text{ from } x \text{ to } z\}.$$

Thus P_x is the path component of x in U .

We first show that P_x is open in U . Let $z \in P_x$. Since U is an open neighborhood of z and X is locally path connected, there exists an open path-connected set V_z such that

$$z \in V_z \subseteq U.$$

For every $w \in V_z$, there is a path from z to w inside V_z . Since $z \in P_x$, there is also a path from x to z inside U . Composing these paths gives a path from x to w , so $w \in P_x$. Hence

$$V_z \subseteq P_x,$$

and therefore P_x is open.

We similarly show that $U \setminus P_x$ is open. Let $z \in U \setminus P_x$. Again choose an open path-connected neighborhood

$$z \in V_z \subseteq U.$$

Suppose that $V_z \cap P_x \neq \emptyset$, and choose $w \in V_z \cap P_x$. There is a path from x to w , and since V_z is path connected, there is a path from w to z . Composing them would give a path from x to z , contradicting $z \notin P_x$. Hence

$$V_z \subseteq U \setminus P_x.$$

Thus $U \setminus P_x$ is also open.

Therefore P_x is both open and closed in U . Since U is connected and P_x is nonempty, we must have

$$P_x = U.$$

Hence every point of U can be joined to x by a path, so U is path connected.

□

Problem 4.3. Let X be a topological space, and let $A, B \subseteq X$ be connected subspaces such that $A \cap B \neq \emptyset$. Prove that $A \cup B$ is connected.

Proof. We are free to discard the trivial case where $A = B$.

Since A, B are connected, the only continuous functions $f: A \rightarrow \{0, 1\}$ and $g: B \rightarrow \{0, 1\}$ are constant functions.

Suppose $A \cup B$ is not connected, meaning we can find a continuous $h: A \cup B \rightarrow \{0, 1\}$ which is non-constant.

Necessarily,

$$h|_A = c_A \in \{0, 1\}$$

is constant, and

$$h|_B = c_B \in \{0, 1\}$$

is constant, since they are connected spaces.

We assume h is non-constant. So there exists some $x, y \in A \cup B$ such that

$$h(x) \neq h(y).$$

Then, necessarily, these points cannot be in the intersection of A and B . So one of these points are in $A \setminus B$ or $B \setminus A$. Without loss of generality, suppose $x \in A \setminus B$. Then we require $y \in B \setminus A$ since otherwise we would have

$$c_A = h(x) \neq h(y) = c_A.$$

From this we deduce $c_A \neq c_B$, but then, for $z \in A \cap B$, we would have

$$h(z) = h|_A(z) = h|_B(z)$$

so

$$c_A = c_B,$$

a contradiction. □

Problem 4.4. Let $\{A_\alpha\}_{\alpha \in I}$ be a collection of connected subspaces of X . Suppose there exists a point $p \in X$ such that $p \in A_\alpha$ for every $\alpha \in I$.

Prove that

$$\bigcup_{\alpha \in I} A_\alpha$$

is connected.

Proof. We suppose A_α are all connected, so for any continuous $f_\alpha : A_\alpha \rightarrow \{0,1\}$, the function must be constant.

Let $\varphi : \bigcup_{\alpha} A_\alpha \rightarrow \{0,1\}$ be continuous.

Then, necessarily,

$$\varphi|_{A_\alpha} : A_\alpha \rightarrow \{0,1\}$$

is constant (also the restriction remains continuous).

We have

$$\varphi(p) = c \in \{0,1\}$$

and in particular, since $p \in A_\alpha$ for every $\alpha \in I$, and the restriction to A_α is constant, we have

$$\text{im } \varphi|_{A_\alpha} = \{c\}$$

for every $\alpha \in I$. Thus

$$\text{im } \varphi = \{c\}$$

is constant, and hence

$$\bigcup_{\alpha} A_\alpha$$

is connected. □

Problem 4.5. Let $A \subseteq X$ be connected. Prove that its closure $\overline{A}$ is connected.

Proof. Suppose A is a connected subspace of some space X . Then any continuous function $f: A \rightarrow \{0,1\}$ must be constant.

Recall that $\overline{A}$ is the smallest closed set containing A , equivalently it is the intersection of all closed subsets of X containing A . Denote $f(A) = \{c_A\}$.

Let $f: \overline{A} \rightarrow \{0,1\}$ be some continuous function.

Suppose, $x, y \in \overline{A}$ with

$$f(x) = c_x, \quad f(y) = c_y.$$

The fiber of c_x is an open set V_x of $\overline{A}$ containing x . Likewise, the fiber of c_y is an open set V_y of $\overline{A}$ containing y .

Then $f(V_x \cap A) = \{c_x\}$ and $f(V_y \cap A) = \{c_y\}$, but since these are subsets of A , and

$$\text{im } f|_A = \{c_A\},$$

we must have

$$c_x = c_A = c_y.$$

Therefore the space is connected.

Note: I should have been more rigorous in explaining that we can do this by arguing that since the points were in the closure, any open neighborhood necessarily intersects A .

□

Problem 4.6. Let C be a connected component of a topological space X . Prove that C is closed in X .
You may use A connected $\Rightarrow \overline{A}$ connected.

Proof. C is a maximal connected subset.

Thus there does not exist any $D \subseteq X$ containing C which is also connected.

In particular, since $C \subseteq \overline{C}$ we must have that

$$C = \overline{C}.$$

Otherwise, $\overline{C}$ would be a larger connected subspace (C connected $\Rightarrow \overline{C}$ connected) containing all of C , contradicting the maximality of C . □

Problem 4.7. Let X be locally connected. Let $U \subseteq X$ be open, and let C be a connected component of U . Prove that C is open in X .

Proof. Since X is locally connected, then for every $x \in X$ we can, for every open neighborhood $U_x \supseteq x$, find an open connected set $V_x \supseteq x$ contained in U_x .

We suppose U is an open set in X and that C is a maximal connected subset of U .

For any $x \in C$, we can find a connected open neighborhood V_x with $V_x \subseteq U$.

Since C is maximal, $x \in C$, and V_x is connected we have $V_x \subseteq C$ for every x (might need one more step to be completely justified). Then

$$\bigcup_x V_x = C.$$

Since open sets are closed under arbitrary unions, C is open.

□

5 §27 and 29

In $\mathbb{R}$, we can study compactness in a more familiar setting (more Real Analysis, damn it).

We recall, from Real Analysis, that in $\mathbb{R}$, compactness corresponds to being closed and bounded.

However, this equivalence is special to Euclidian-like spaces. We have seen that this does not generally hold.

5.1 Closed intervals are compact

The first major result is that

$$[a, b] \subseteq \mathbb{R}$$

is compact, always.

Essentially, this is just Heine-Borel in one dimension. Munkres uses the least upper bound property of $\mathbb{R}$.

Suppose $\mathcal{U}$ is some open cover of $[a, b]$.

Define

$$C = \{x \in [a, b] \mid [a, x] \text{ can be covered by finitely many members of } \mathcal{U}\}.$$

The strategy is to show that this set is nonempty, let $c = \sup C$, and prove that $c = b$.

Suppose instead $c < b$. Since $\mathcal{U}$ covers $[a, b]$, some $U \in \mathcal{U}$ contains c .

Because U is open it contains an interval $(c - \varepsilon, c + \varepsilon)$.

Since $c = \sup C$, there is some $x \in C$ with $c - \varepsilon < x \leq c$.

Now $[a, b]$ already has a finite subcover, and U extends that finite cover beyond c .

So some point larger than c belongs to C , a contradiction. So we conclude $c = b$, so the interval has a finite subcover.

5.2 Heine-Borel (again)

From the previous result, together with the general compactness results from §26, we get that

$$A \subseteq \mathbb{R}^n \text{ is compact} \Leftrightarrow A \text{ is closed and bounded.}$$

Proof sketch. $\Rightarrow$.

Since $\mathbb{R}^n$ is Hausdorff, A compact implies that A is closed.

For unboundedness, suppose A were unbounded. Then

$$\{B(0, n) \mid n \in \mathbb{N}\}$$

is an open cover of A .

Any finite subcollection is contained in one largest ball

$$B(0, N),$$

which cannot cover unbounded A .

Contradiction, meaning A is bounded.

$\Leftarrow$.

If $A \subseteq \mathbb{R}^n$ is bounded, then for some M ,

$$A \subseteq [-M, M]^n.$$

The box is compact. Since A is closed in $\mathbb{R}^n$ it is closed and compact in the box. Closed subsets of compact spaces are compact so compactness of A follows. $\square$

5.3 Extreme Value Theorem

Suppose X is compact and $f: X \rightarrow \mathbb{R}$ is continuous.

Then $f(X)$ has a maximum and minimum. So there exists $x_{\min}, x_{\max} \in X$ such that $f(x_{\min}) \leq f(x) \leq f(x_{\max})$ for all $x \in X$.

So what we learned in calculus,

continuous function on closed bounded interval attains max/min
is generalized to

continuous images preserve compactness.

5.4 Uniform continuity

Recall that continuity in Real Analysis meant

$$\forall x, \forall \varepsilon > 0, \exists \delta_x > 0 : d(x, y) < \delta_x \Rightarrow d(f(x), f(y)) < \varepsilon$$

with the distinction that δ_x may depend on x .

Recall that uniform continuity requires that for every $\varepsilon > 0$ we can find some point-independent $\delta > 0$ such that the condition holds.

Theorem 5.1. *If X is compact and $f : X \rightarrow Y$ is continuous between metric spaces, then f is uniformly continuous.*

Intuitively this makes sense, since the compactness allows us to have finitely many local choices of δ for which we can simply take the minimum.

5.5 Distance from a point to a set

We define $d(x, A) = \inf\{d(x, a) \mid a \in A\}$.

If A is compact, the infimum is achieved, meaning there is actually some $a_0 \in A$ such that $d(x, A) = d(x, a_0)$.

With the previous fact this is trivial to prove, as $a \mapsto d(x, a)$ is continuous on compact A , hence attains its minimum.

5.6 Closed and bounded does not imply compact in general

In $\mathbb{R}^n$, compact coincides precisely with being closed and bounded, but in arbitrary metric spaces this fails.

For instance, an infinite set with the discrete metric $d(x, y) = 0$ whenever $x = y$ and 1 otherwise, is nicely bounded and the whole space is closed. However, the singleton cover $\{\{x\} \mid x \in X\}$ has no finite subcover, meaning X is not compact.

6 §29 Local compactness

Definition 6.1. A space X is locally compact at x if there exists a compact subspace $C \subseteq X$ that contains a neighborhood of x . Equivalently, there exists an open neighborhood U of x such that $x \in U \subseteq C$ for some compact C . If it holds for every $x \in X$ we say the space X is locally compact.

Take for example $\mathbb{R}$. This space is not compact, but it is obviously locally compact since every point x lies in

$$(x-1, x+1) \subseteq [x-1, x+1],$$

and the closed interval is compact.

Similarly, $\mathbb{R}^n$ is not compact, though it is locally compact.

6.1 Compact implies locally compact

Unsurprisingly, compactness of X implies that every point $x \in X$ has the neighborhood X contained in the compact set X . Not terribly complicated.

The converse fails. Also not very surprising.

6.2 $\mathbb{Q}$ is not locally compact

In a useful contrast with the reals, the rationals is not locally compact.

Take $x \in \mathbb{Q}$.

Any neighborhood in $\mathbb{Q}$ contains something like $(a, b) \cap \mathbb{Q}$. Its closure is roughly $[a, b] \cap \mathbb{Q}$, but this is not compact.

One way to see why is that there are rational sequences converging in $\mathbb{R}$ to irrational numbers.

For instance, $q_n \rightarrow \sqrt{2}$ with all $q_n \in \mathbb{Q}$.

Such a sequence has no convergent subsequence in the rationals with a limit in the rationals. So $\mathbb{Q}$ has no compact neighborhood around a point.

6.3 Stronger local form in Hausdorff spaces

Munkres proves that if X is Hausdorff, then local compactness is equivalent to the following more natural statement:

For every $x \in X$ and every neighborhood U of x , there exists a neighborhood V of x such that $\overline{V}$ is compact, and

$$\overline{V} \subseteq U.$$

As per usual, Hausdorff spaces are nicely behaved.

In particular, we can find arbitrarily small relatively compact neighborhoods.

6.4 One-point compactification

Suppose X is locally compact Hausdorff, but not compact.

Add a new point

$$X^* = X \cup \{\infty\}.$$

How should neighborhood of ∞ look?

The idea will be that being close to ∞ means being outside of some compact part of X .

So define neighborhood of ∞ by sets of the form $X^* \setminus K$, where $K \subseteq X$ is compact.

Together with the original open sets of X , these generate a topology on X^* .

Then X^* is compact Hausdorff.

This is called the one-point compactification of X .

It sounds weird, but it makes sense (trust me bro).

Suppose we have an open cover of X^* . Some open set U contains ∞ . By definition, $X^* \setminus U$ is a compact subset $K \subseteq X$.

So U already covers everything except a compact chunk.

The rest of the open cover covers K .

Compactness of K gives finitely many sets covering it.

Add U . Then we are done.

6.5 Some examples and then practice problems

The one-point compactification of $\mathbb{R}$ is homeomorphic to S^1 .

Likewise, $\mathbb{R}^2 \cup \{\infty\}$ is homeomorphic to S^2 and more generally,

$$(\mathbb{R}^n)^* \cong S^n.$$

Problem 6.1. Let X be a compact space and let $f : X \rightarrow \mathbb{R}$ be continuous. Prove that f attains a maximum and minimum. Prove it using only the fact that continuous images of compact spaces are compact and what compact subsets of $\mathbb{R}$ look like.

Proof. We let X be compact and $f : X \rightarrow \mathbb{R}$ be continuous.

Then

$$f(X)$$

is compact. By the Heine-Borel theorem, $f(X)$ is closed and bounded.

Then for any x in the fiber of $\operatorname{argmax} f$, $f(x) \geq f(y)$ for every $y \in X$ and similarly, any x in the fiber of $\operatorname{argmin} f$ satisfies $f(x) \leq f(y)$ for all $y \in X$.

Importantly, $\operatorname{argmax} f = \sup f(X)$ which is finite (from boundedness) and in $f(X)$ since $f(X)$ is closed. Analogously, $\operatorname{argmin} f = \inf f(X)$ which, once again, is finite and in $f(X)$. $\square$

Problem 6.2. Let X be a noncompact locally compact Hausdorff space. Define $X^* = X \cup \{\infty\}$, and declare a set $U \subseteq X^*$ open if either:

- $U \subseteq X$ and U is open in X , or
- $\infty \in U$ and $X^* \setminus U$ is compact in X .

Prove that X^* , the one-point compactification of X , is compact.

Proof. Let $U \subseteq X^*$ be some open set.

Then, recall throughout that we either have that U is open in X or $\infty \in U$ and $X^* \setminus U$ is compact in X .

Let $\mathcal{U} = \{U_\alpha\}_\alpha$ be some cover of X^* ,

$$\bigcup_{\alpha} U_{\alpha} = X^*.$$

Necessarily, one of these U_α must contain ∞ . Then

$$X^* \setminus U_\alpha$$

is compact, meaning $\mathcal{U}$, as a cover of $X^* \setminus U_\alpha$ can be reduced to a finite cover

$$\bigcup_{i=1}^n U_i = X^* \setminus U_\alpha.$$

Then,

$$U_\alpha \cup \bigcup_{i=1}^n U_i = U_\alpha \cup X^* \setminus U_\alpha = X^*.$$

So we have found a finite cover from the general cover, meaning X^* is compact. $\square$

Problem 6.3. Let X be a noncompact locally compact Hausdorff space, and let

$$X^* = X \cup \{\infty\}$$

have the one-point compactification topology. Prove that X^* is Hausdorff.

Proof. X is already Hausdorff so separating two points $x, y \in X$ is not an issue.

Fix $x \in X^* \setminus \{\infty\}$. We will look at separating x and ∞ .

A set $U \subseteq X^*$ containing ∞ is open if and only if $X^* \setminus U$ is compact in X .

Ideally we would want to prove that there must exist some open set U with $x \notin U$ for our arbitrary $x \in X = X^* \setminus \{\infty\}$, for which $X^* \setminus U$ is compact.

Since X is locally compact (and Hausdorff), we can, for any open neighborhood V of x find $W \subseteq V$ containing x such that $\overline{W}$ is compact.

Since X is Hausdorff and $\overline{W}$ is compact, we have that $\overline{W}$ is closed.

Thus $X \setminus \overline{W}$ is open in X . Therefore, $X^* \setminus \overline{W}$ is open in X^* . We don't really need to make this observation, since it suffices that $\overline{W}$ is compact $\Rightarrow X^* \setminus \overline{W}$ is open in X^* .

Furthermore, $\infty \in X^* \setminus \overline{W}$ while, clearly, $x \notin X^* \setminus \overline{W}$.

Since $X^* \setminus \overline{W}$ and $\overline{W}$ is disjoint, and $W \subseteq \overline{W}$, we have that $W \cap X^* \setminus \overline{W} = \emptyset$.

So $X^* \setminus \overline{W}$ is an open neighborhood of ∞ , while W is an open neighborhood of x .

Furthermore, they are disjoint.

As x was arbitrarily chosen, we can separate any $x \in X^* \setminus \{\infty\}$ from ∞ .

This together with the fact that X on its own is Hausdorff, gives Hausdorffness for X^* . $\square$

7 Countability- and Separation Axioms

7.1 Why do we need these axioms?

A topology can be quite large.

For instance, to know that $f: X \rightarrow Y$ is continuous, in principle we have to understand all open sets in Y . To understand what happens near x , we might have to consider all neighborhoods of x .

Countability axioms let us ask whether we can replace this potentially gigantic amount of information by a countable amount.

There are two fundamentally different notions, those being

first countable = countably manageable locally

and

second countable = countably manageable globally.

7.2 A basis at a point

Before first countability, we distinguish two concepts.

A basis for the topology is a family $\mathcal{B}$ such that every open set is a union of elements of $\mathcal{B}$, as we already know.

A **basis at a point** x , or local basis at x , is a collection $\mathcal{B}_x$ of neighborhoods of x such that for every neighborhood U of x , there exists

$$B \in \mathcal{B}_x$$

with

$$x \in B \subseteq U.$$

We can think of this as being arbitrarily close to x , as we often pondered in analysis.

7.3 First countability

A space X is **first countable** if every point $x \in X$ has a countable local basis.

Thus for each x ,

$$\mathcal{B}_x = \{B_1, B_2, \dots\}$$

is enough to probe every neighborhood of x .

Every metric space is first countable, perhaps unsurprisingly.

Indeed, at x take

$$\mathcal{B}_x = \{B_{1/n}(x) \mid n \in \mathbb{N}\}.$$

Given any neighborhood $U \ni x$, because U is metric-open there is some $\varepsilon > 0$ with

$$B_\varepsilon(x) \subseteq U.$$

Choose n sufficiently large that

$$\frac{1}{n} < \varepsilon.$$

Then

$$B_{1/n}(x) \subseteq B_\varepsilon \subseteq U.$$

So metric spaces are first countable.

7.4 Sequences

Recall that $x \in \overline{A}$ means every neighborhood of x intersects A .

Suppose there is a sequence

$$a_n \in A$$

with

$$a_n \rightarrow x.$$

Then necessarily

$$x \in \overline{A}.$$

This is because every neighborhood U of x contains all sufficiently large a_n , hence intersects A .

That direction holds in every topological space

$$a_n \in A, a_n \rightarrow x \Rightarrow x \in \overline{A}.$$

However, the converse fails in arbitrary topological spaces.

First countability, however, makes it hold.

7.5 The sequence lemma

If X is first countable, then

$$x \in \overline{A} \Leftrightarrow \exists a_n \in A, a_n \rightarrow x.$$

Let

$$B_1, B_2, \dots$$

be a countable local basis at x .

It is convenient to make it nested,

$$C_n = \bigcap_{i \in [n]} B_i.$$

Then

$$C_1 \supseteq C_2 \supseteq C_3 \supseteq \dots$$

and $\{C_n\}$ is still a local basis.

Since $x \in \overline{A}$, every C_n intersects A . Choose $a_n \in C_n \cap A$. Now let U be any neighborhood of x . Since $\{C_n\}$ is a local basis, some

$$C_N \subseteq U.$$

For $n \geq N$,

$$a_n \in C_n \subseteq C_N \subseteq U.$$

Therefore

$$a_n \rightarrow x.$$

7.6 Continuity can also be tested by sequences

Recall from analysis, that

$$f \text{ continuous} \Rightarrow x_n \rightarrow x \Rightarrow f(x_n) \rightarrow f(x).$$

That implication is true in arbitrary topological spaces.

But once again, the converse generally fails.

If the domain X is first countable, the converse holds.

So all the sequential reasoning from analysis works so smoothly largely because metric spaces are first countable.

7.7 Second countability

Now we make countability global.

A space X is **second countable** if its topology has a countable basis,

$$\mathcal{B} = \{B_1, B_2, \dots\}.$$

This is substantially stronger than first countability, and obviously implies first countability.

It should not be terribly surprising that the converse fails.

7.8 $\mathbb{R}$ is second countable

The usual topology on $\mathbb{R}$ has the basis

$$\mathcal{B} = \{(p, q) \mid p, q \in \mathbb{Q}, p < q\}.$$

Since $\mathbb{Q} \times \mathbb{Q}$ is countable, the basis $\mathcal{B}$ is countable.

7.9 Two or more countability properties

A space X is **separable** if it has a countable dense subset

$$\exists D = \{d_1, d_2, \dots\} \subseteq X$$

such that

$$\overline{D} = X.$$

Intuitively, countably many points come arbitrarily close to every point of the space.

For instance $\mathbb{Q}$ is dense in $\mathbb{R}$, so $\mathbb{R}$ is separable.

A space X is **Lindelöf** if every open cover has a countable subcover.

Comparatively,

compact = every open cover has a finite subcover,

while

Lindelöf = every open cover has a countable subcover.

Thus

compact $\Rightarrow$ Lindelöf.

7.10 The central implication picture

Second countability is very strong, as it implies first countable, separable and Lindelöf.

The latter two are worth understanding.

Second countable $\Rightarrow$ separable.

Let

$$\mathcal{B} = \{B_1, B_2, \dots\}$$

be a countable basis.

For every nonempty B_n , choose

$$x_n \in B_n.$$

Let

$$D = \{x_n\}.$$

Then D is countable.

If $U \subseteq X$ is nonempty open, U contains some basis element B_n , and therefore

$$x_n \in U \cap D.$$

So every nonempty open set intersects D .

Hence

$$\overline{D} = X.$$

Thus X is separable.

Second countable $\Rightarrow$ Lindelöf.

Let $\mathcal{U}$ be an open cover.

For each basis element B_n that is contained in some member of $\mathcal{U}$, choose one such member

$$U_n \in \mathcal{U}.$$

There are only countably many B_n , hence only countably many selected U_n .

These cover X . If $x \in X$, choose some $U \in \mathcal{U}$ containing x . Since $\mathcal{B}$ is a basis, there exists

$$x \in B_n \subseteq U.$$

So the selected U_n corresponding to B_n contains x .

Therefore X is Lindelöf.

7.11 In metric spaces, many of these notions collapse

For a metric space,

$$\text{second countable} \Leftrightarrow \text{separable} \Leftrightarrow \text{Lindelöf}.$$

First countability is automatic.

One useful direction is $\text{separable} \Rightarrow \text{second countable}$.

Suppose

$$D = \{d_1, d_2, \dots\}$$

is countable and dense.

Consider all balls

$$B_r(d), \quad d \in D, \quad r \in \mathbb{Q}_{>0}.$$

There are countably many and they form a basis.

Note: Separation axioms next.